Biomolecular Condensates and the Origin of Life

From Flory–Huggins thermodynamics to protocellular compartments — first-principles mathematical modelling of liquid–liquid phase separation in biology

Wolfram Community submission — June 2026

Abstract

Biomolecular condensates are **membrane-free liquid droplets** inside cells, formed by liquid–liquid phase separation (LLPS) of proteins and nucleic acids. Discovered as P granules in *C. elegans* germ cells (Brangwynne et al., 2009), they have since been found to organise virtually every aspect of cellular function: gene regulation, signal transduction, stress response, and the formation of structures from the nucleolus to synaptic densities.

This notebook develops the mathematical physics of condensates from first principles. We implement the **Flory–Huggins** free energy to derive the phase diagram, solve the **Cahn–Hilliard** partial differential equation for spinodal decomposition, compute the **concentration enhancement** that makes condensates natural chemical reactors, and model the **protocell dynamics** (growth, division, competition) that connect condensates to the origin of life. We conclude with the **liquid → gel → aggregate** transition that links condensate dysfunction to neurodegenerative disease.

Audience: the prose is written for two overlapping audiences. Each section opens with an intuitive explanation that requires no background, then deepens into a quantitative treatment with equations, numerical integration, and visualisations. Biophysicists will recognise the canonical references; readers from other backgrounds should be able to follow the physical narrative.

1. The membrane-free organelle

For over a century, cell biology was built on the assumption that the cell's internal organisation depends on **membranes**: lipid bilayers that enclose the nucleus, the mitochondria, the endoplasmic reticulum. Compartmentalisation by membranes was the organising principle of eukaryotic life.

Then, in 2009, **Brangwynne, Eckmann, Courson, Rybarska, Hoege, Gharakhani, Jülicher, and Hyman** showed that P granules — RNA–protein bodies in the germ cells of the nematode *C. elegans* — are not membrane-bound organelles at all. They are **liquid droplets**. They flow, they fuse, they drip, they dissolve and re-form. They are phase-separated condensates of RNA and proteins, held together not by a lipid membrane but by the same physics that separates oil from water.

This discovery launched a revolution. Within a decade, dozens of cellular structures were reclassified as condensates: stress granules, nucleoli, Cajal bodies, PML bodies, centrosomes, synaptic densities, heterochromatin domains, transcriptional hubs. The 2024 Nobel Prize in Chemistry was awarded to David Baker for protein design, but the intellectual context — that proteins are not just rigid machines but can phase-separate into functional liquids — owes as much to the condensate revolution.

The physics. The formation of a condensate is a thermodynamic phase transition. Below a critical temperature (or above a critical interaction strength), a homogeneous solution of macromolecules becomes thermodynamically unstable and spontaneously separates into a dilute phase and a dense phase. The dense phase is the condensate. The transition is governed by a competition between the entropy of mixing (which favours homogeneity) and the enthalpic interactions between molecules (which favour like-with-like aggregation). The mathematical framework for this competition is **Flory–Huggins theory**, which we derive in the next section.

The origin-of-life connection. Perhaps the most provocative implication of the condensate revolution is its connection to the **origin of life**. Aleksandr Oparin proposed in 1924 that life began inside **coacervate droplets** — spontaneously forming colloidal structures that concentrate organic molecules from the dilute primordial ocean. Oparin's hypothesis was dismissed for decades because the known coacervates were too disordered to sustain biochemistry. But modern condensates, with their exquisite sequence-dependent phase behaviour and their ability to concentrate specific enzymes by factors of 100–1000, revive Oparin's vision in molecular detail. We show in §4 that condensate-assisted chemistry solves the **concentration problem** that has bedevilled prebiotic chemistry: how do you get reactions going in a picomolar ocean?

2. Thermodynamics of phase separation

2.1 The Flory–Huggins free energy

The lattice model. Consider a lattice of N_t sites, each occupied by either a monomer of species 1 (the polymer, chain length N_1) or a monomer of species 2 (the solvent, with $N_2 = 1$). Let ϕ be the volume fraction of species 1. The Flory–Huggins free energy of mixing per lattice site is:

$$\frac{f}{\mathrm{kT}} = \frac{\phi \ln \phi}{N_1} + \frac{(1-\phi)\ln(1-\phi)}{N_2} + \chi\,\phi(1-\phi)$$

The first two terms are the **entropy of mixing** — always negative (favour mixing) and scaled by the inverse chain lengths $1/N_1$ and $1/N_2$. Longer chains have less translational entropy per monomer, so polymers phase-separate more readily than small molecules. The third term is the **enthalpy of mixing**, parameterised by the Flory interaction parameter χ. For most condensate-forming proteins, $\chi > 0$ (unfavourable mixing), and χ decreases with temperature ($\chi \propto 1/T$), giving an upper critical solution temperature (UCST) phase diagram.

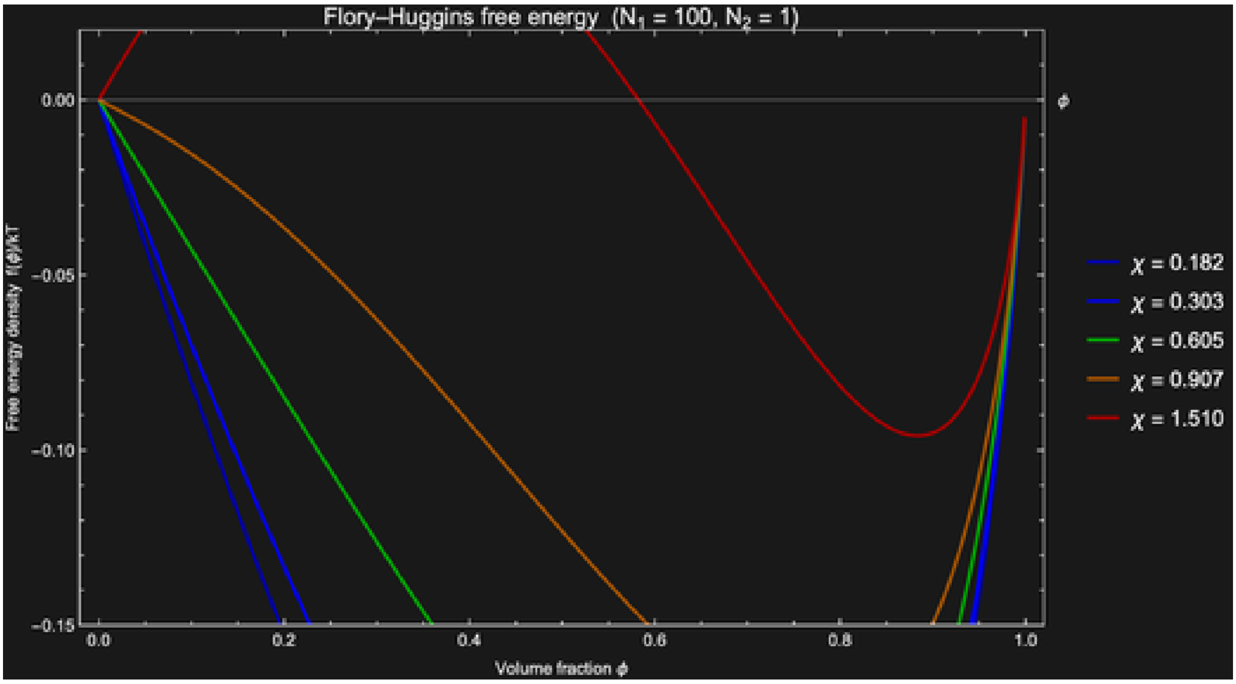

Figure 2.1 Flory–Huggins free energy density for a polymer–solvent system ($N_1 = 100$, $N_2 = 1$) at several values of χ. Below the critical χ, the free energy has a single minimum (stable mixed state). Above it, a double well appears: the system phase-separates into dilute (ϕ_l) and dense (ϕ^d) phases at the two minima.

2.2 Spinodal and binodal: the phase diagram

The thermodynamic stability of the mixed state is determined by the curvature of the free energy. The **spinodal** is the locus of inflection points:

$$\frac{\partial^2 f}{\partial \phi^2} = \frac{1}{N_1\,\phi} + \frac{1}{N_2\,(1-\phi)} - 2\,\chi = 0$$

Inside the spinodal, the free energy is concave ($\partial^2 f/\partial\phi^2 < 0$) and the system is **unstable** to infinitesimal composition fluctuations — this is **spinodal decomposition**. Between the spinodal and the **binodal** (the coexistence curve), the system is **metastable** — it requires a finite-amplitude fluctuation (nucleation) to phase-separate.

The binodal is found by the **common tangent construction**: the two coexisting compositions ϕ_l and ϕ^d satisfy equal chemical potential and equal osmotic pressure:

$$\mu(\phi_L) = \mu(\phi_R), \quad \Pi(\phi_L) = \Pi(\phi_R)$$

The **critical point** is where the spinodal and binodal meet:

$$\phi_c = \frac{\sqrt{N_2}}{\sqrt{N_1} + \sqrt{N_2}}, \quad \chi_c = \frac{\left(\frac{1}{\sqrt{N_1}} + \frac{1}{\sqrt{N_2}}\right)^2}{2}$$

For a long polymer ($N_1 = 100, N_2 = 1$): $\phi^c = 0.091, \chi^c = 0.605$. The critical concentration is low (~9%), meaning that even dilute polymer solutions phase-separate when χ exceeds a modest threshold.

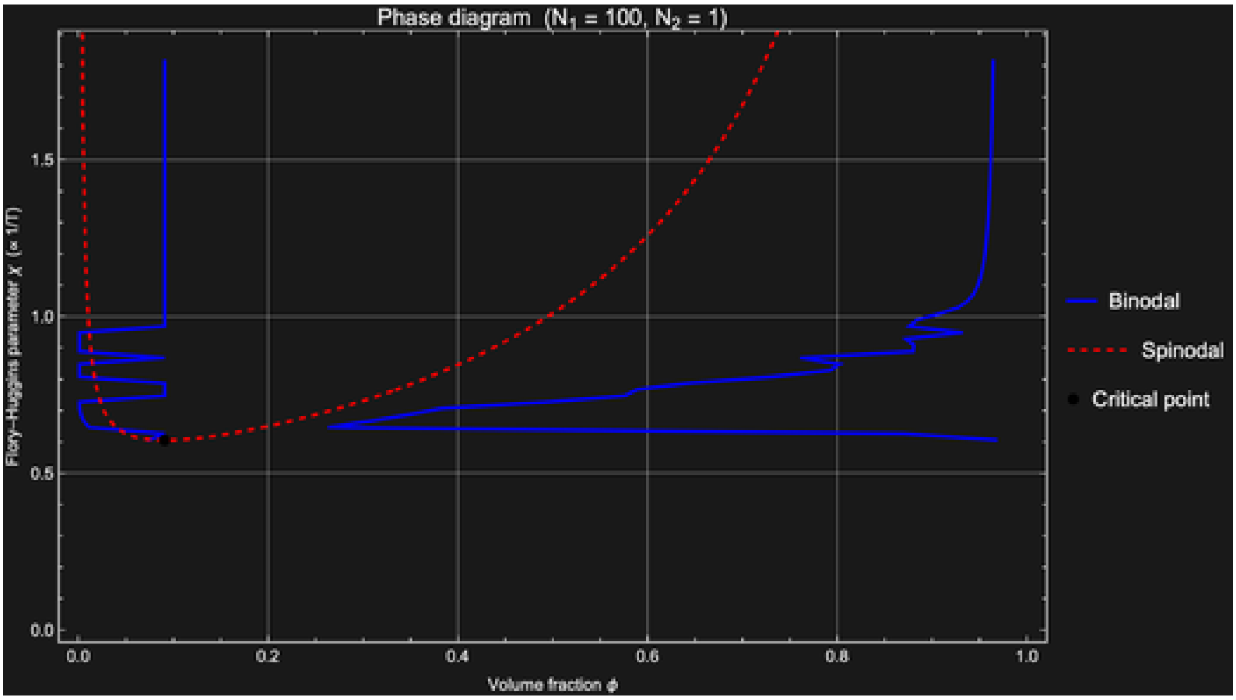

Figure 2.2 Phase diagram in χ–ϕ coordinates. The blue curve is the binodal (coexistence); the red dashed curve is the spinodal (stability limit). Between them: metastable (nucleation and growth). Inside the spinodal: unstable (spinodal decomposition). The black dot is the critical point.

2.3 Chain length and multivalency

A key prediction of Flory–Huggins theory is that **longer polymers phase-separate more readily**. The critical χ decreases as $1/\sqrt{N}$, meaning that a polymer of 1000 monomers needs only 1/10th the interaction strength of a 10-mer. This is biologically significant: intrinsically disordered regions (IDRs) of condensate-forming proteins are typically 50–500 residues, and their effective N is further increased by multivalent interactions (multiple binding sites per molecule, RNA scaffolds, post-translational modifications).

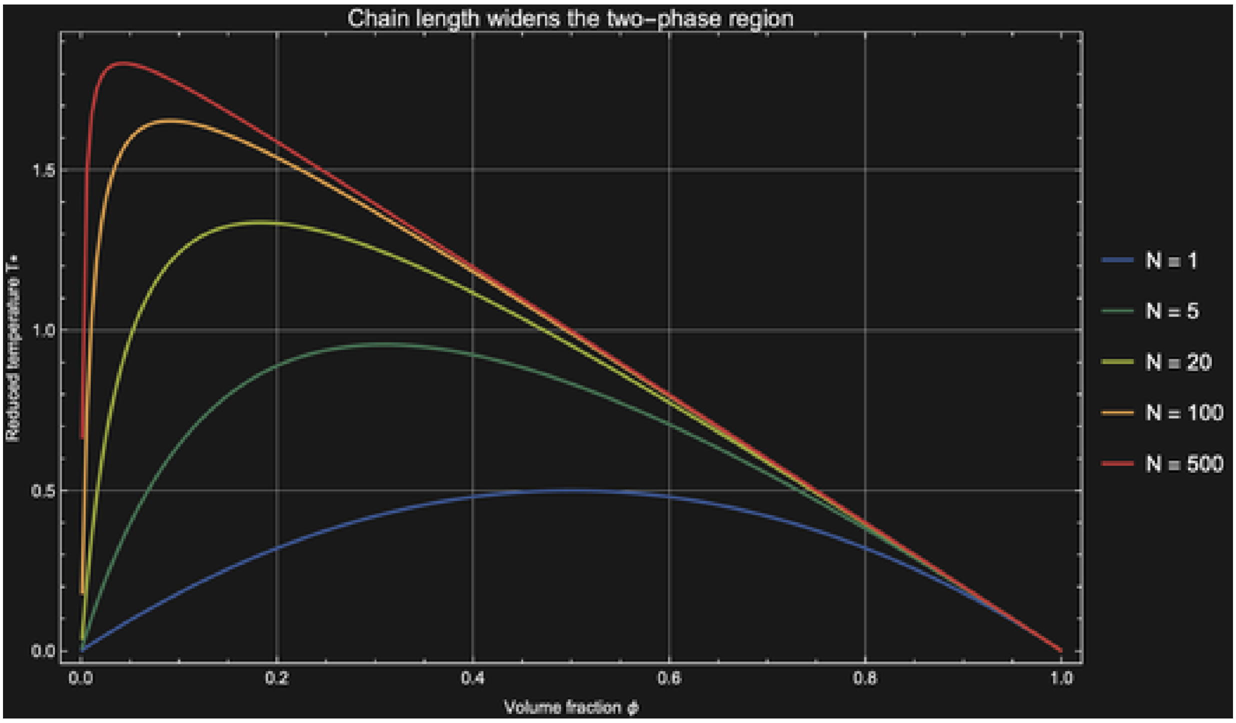

Figure 2.3 Spinodal curves for different chain lengths. Longer chains produce wider two-phase regions at higher temperatures (lower χ threshold). This is why proteins with long disordered regions are prone to phase separation.

3. Cahn–Hilliard dynamics of droplet formation

3.1 The equation

The Flory–Huggins theory tells us *whether* phase separation occurs, but not *how*. The dynamics of a conserved order parameter (composition must be conserved — matter doesn't appear or disappear) are governed by the **Cahn–Hilliard equation**:

$$\frac{\partial \phi}{\partial t} = M \nabla^2 \left(\frac{\delta F}{\delta \phi} - \kappa \nabla^2 \phi \right)$$

where M is the mobility (diffusion coefficient), $\delta F/\delta\phi$ is the chemical potential from the Flory–Huggins free energy, and κ is the gradient energy coefficient that penalises sharp interfaces (it sets the surface tension γ of the condensate).

Physical content. The Cahn–Hilliard equation says: material flows down chemical potential gradients (as in ordinary diffusion), but the chemical potential now includes a penalty for composition gradients (the κ term). Inside the spinodal region, the bulk chemical potential drives **uphill diffusion** — material flows from dilute to concentrated regions, amplifying any fluctuation. The gradient penalty prevents infinitely sharp interfaces and selects a characteristic wavelength for the initial instability.

3.2 Spinodal decomposition

We solve the 1D Cahn–Hilliard equation using Wolfram's built-in `NDSolve` with the Method of Lines. The initial condition is the mean composition $\phi = 0.3$ plus small sinusoidal perturbations.

```
sol = NDSolve[{
   D[ϕ[x, t], t] == mob D[
      mu[ϕ[x, t]] - κ D[ϕ[x, t], x, x], x, x],
   ϕ[x, 0] == ϕ0 + noise Sin[2π k x / L],
   ϕ[0, t] == ϕ[L, t],  (* periodic BCs *)
   ϕ'[0, t] == ϕ'[L, t]},
   ϕ, {x, 0, L}, {t, 0, 50},
   Method -> {"MethodOfLines",
     "SpatialDiscretization" -> {"TensorProductGrid",
       "MinPoints" -> 200}}];
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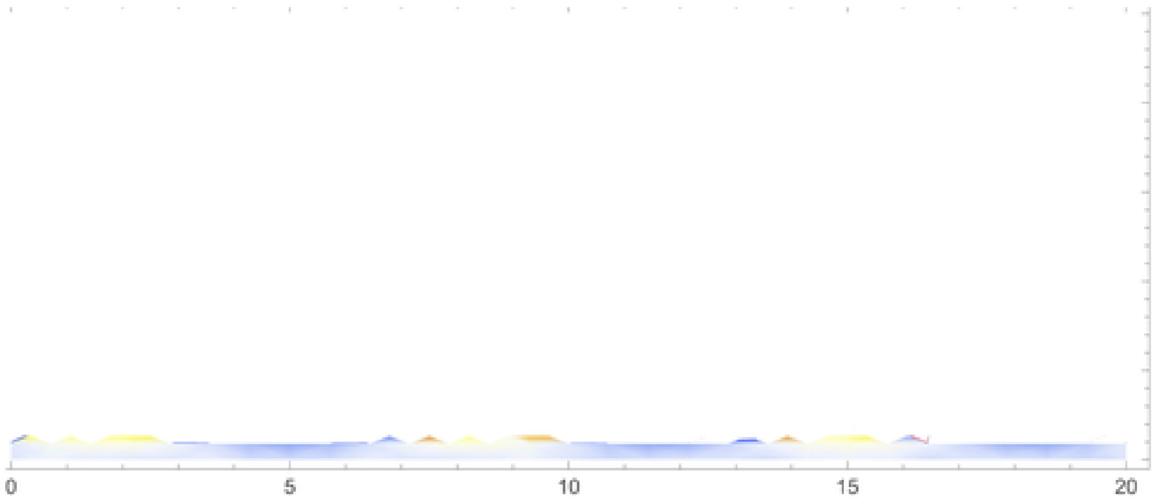


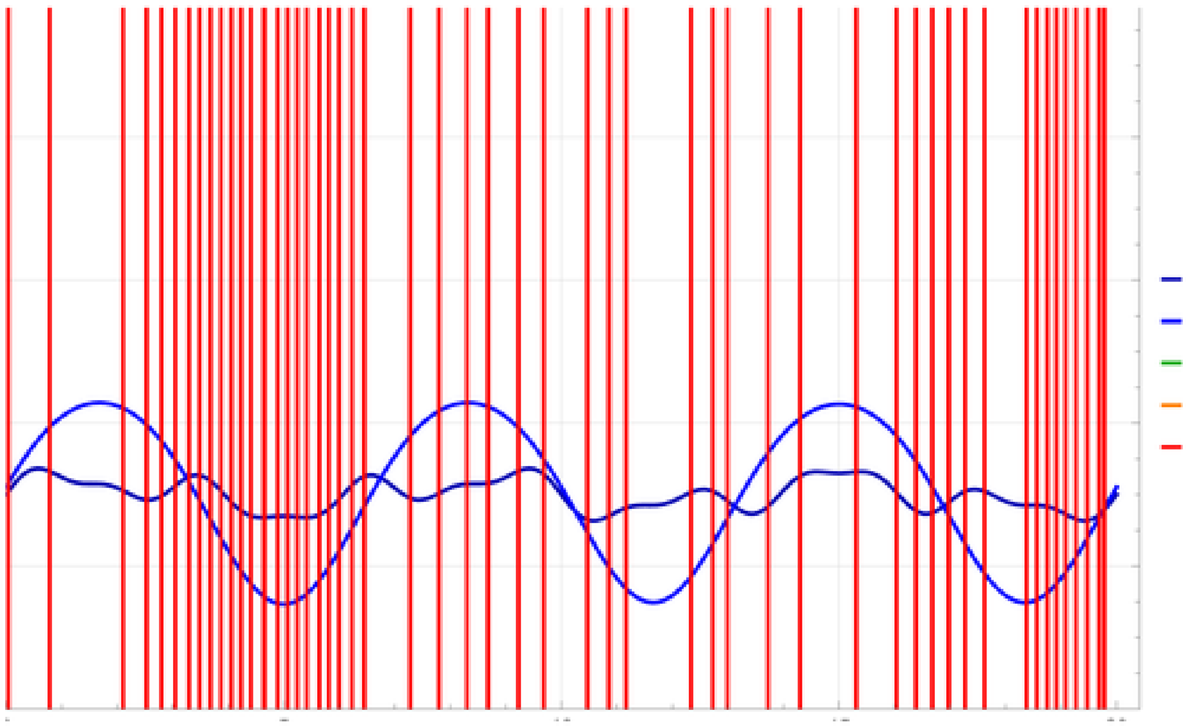


Figure 3.1 Top: space–time density plot of the 1D spinodal decomposition. The initially uniform composition rapidly develops concentration fluctuations that coarsen into well-separated dense (red) and dilute (blue) domains. Bottom: composition profiles at selected times showing the progression from small perturbation to fully phase-separated condensates.

3.3 Coarsening: Lifshitz–Slyozov–Wagner

Once phase separation is complete, the system continues to evolve: large droplets grow at the expense of small ones. This is **Ostwald ripening**, described by the Lifshitz-Slyozov–Wagner (LSW) theory. The critical radius $R*$ is set by the Gibbs–Thomson equation:

$$c_{\text{eq}}\left(R\right) = c_{\infty} \exp\left(\frac{2\,\gamma\,V_m}{R\,k\,T}\right) \approx c_{\infty}\left(1 + \frac{l_c}{R}\right)$$

Droplets with $R < R*$ have higher equilibrium concentration than the bulk — they dissolve. Droplets with $R > R*$ have lower equilibrium concentration — they grow. The mean domain size grows as $\langle R\rangle \propto t$^$(1/3)$, the classic LSW scaling.

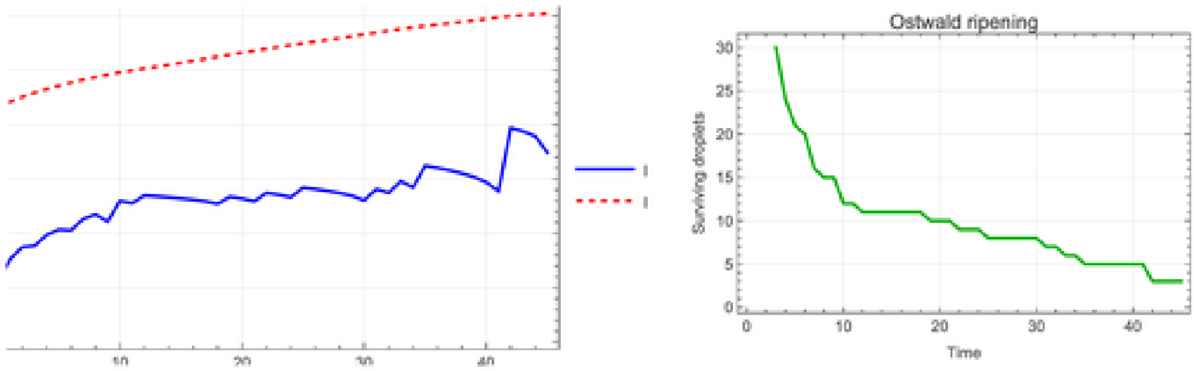


Figure 3.2 Ostwald ripening of a polydisperse condensate population. Upper left: size distribution evolution (early: broad, late: narrower with larger mean). Upper right: number of surviving droplets decreases as small ones dissolve. Lower left: mean and max radius vs time with t^(1/3) fit. Lower right: schematic showing material transfer from small to large droplets.

4. Concentration enhancement and catalysis

4.1 The partition coefficient

When a condensate forms, molecules partition between the dense and dilute phases according to their chemical affinity. The **partition coefficient** K is defined as:

$$K = \frac{c_{\text{in}}}{c_{\text{out}}}$$

For condensate-forming proteins, K is typically 10–100. For RNA molecules that interact strongly with the condensate scaffold, K can reach 100–1000. A molecule at 1 nM bulk concentration with K = 100 is at 100 nM inside the condensate. A molecule at 1 nM with K = 1000 is at 1 μM. This is the concentration miracle.

4.2 Bimolecular rate enhancement

For a bimolecular reaction A + B → C with second-order rate constant k:

$$\text{rate}_{\text{dilute}} = k\,[A]_{\text{out}}\,[B]_{\text{out}}$$

$$\text{rate}_{\text{condensate}} = k\,K_A\,[A]_{\text{out}}\,K_B\,[B]_{\text{out}} = K_A\,K_B\,\text{rate}_{\text{dilute}}$$

If both reactants partition equally ($K_a \approx K_e \approx K$), the reaction rate is enhanced by **K^2**. With K = 100, this is a **10,000-fold enhancement**. This is the quantitative answer to the concentration problem in prebiotic chemistry: you don't need a concentrated primordial soup if you have phase separation.

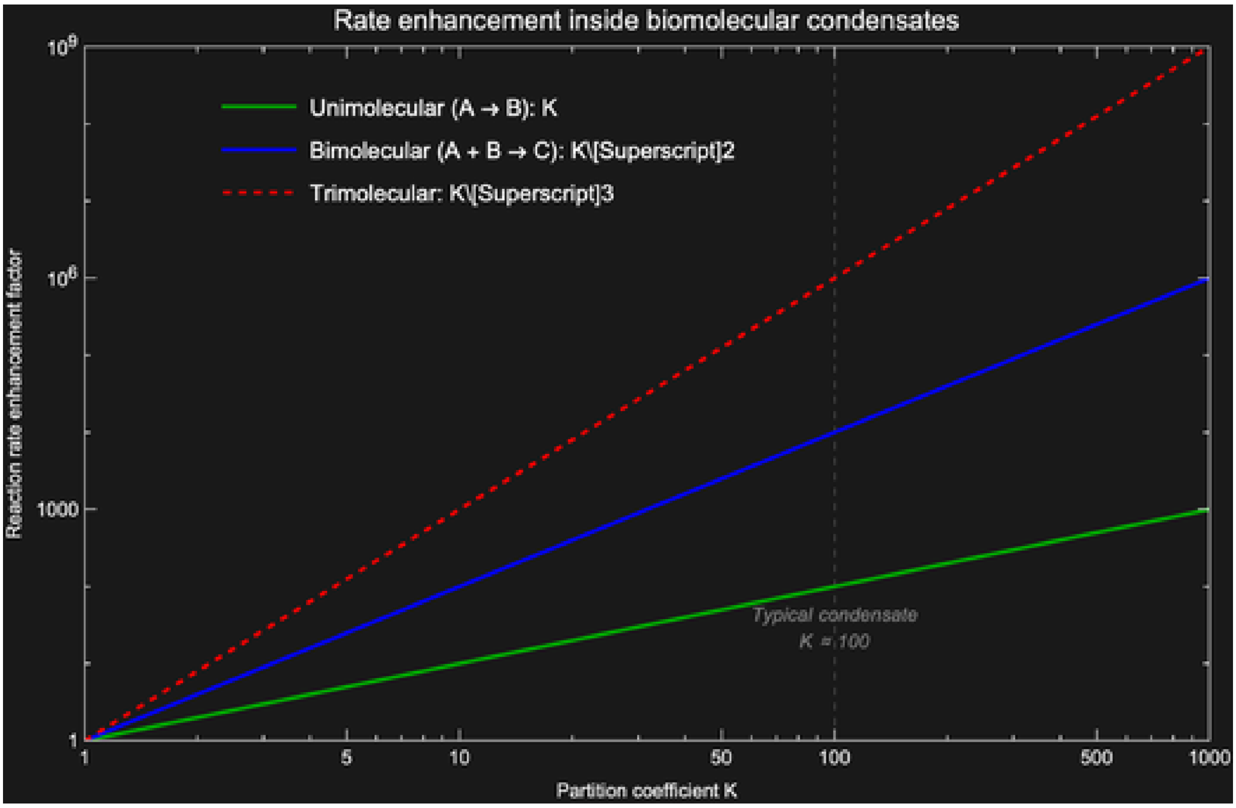


Figure 4.1 Reaction rate enhancement inside condensates as a function of the partition coefficient K. Bimolecular reactions (blue) are enhanced by K^2; trimolecular reactions (red) by K^3. At the typical condensate K ≈ 100, bimolecular rates are 10^4 higher inside the condensate.

4.3 Prebiotic chemistry phase space

Which combinations of partition coefficient and bulk concentration allow net chemical production? We map the net replication rate (replication minus degradation) across the K–c parameter space:

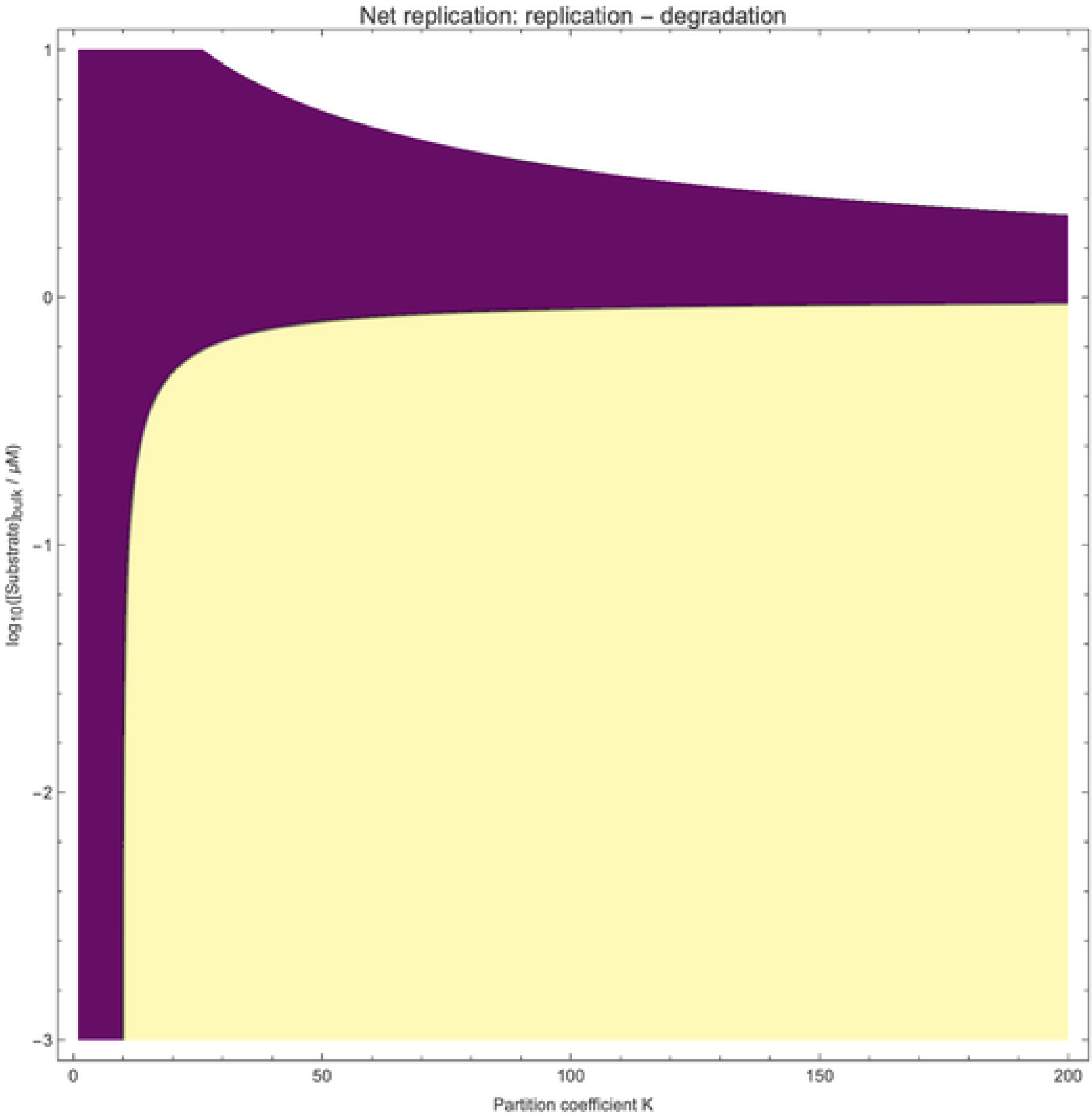


Figure 4.2 Net replication rate as a function of partition coefficient K and bulk substrate concentration. The black contour marks the boundary between net growth (blue) and net decay (red). At low bulk concentrations (the likely prebiotic regime), only condensates with K > 50–100 support net replication.

5. RNA world inside condensates

5.1 Ribozyme kinetics

The RNA world hypothesis proposes that the first self-replicating molecules were RNA ribozymes — RNA molecules that catalyse their own replication. The kinetics of ribozyme-catalysed replication follow **Michaelis–Menten** saturation:

$$v = \frac{V_{\max}\,[S]}{K_m + [S]}$$

Inside a condensate, both the enzyme (ribozyme) and the substrate (template RNA) are concentrated by the partition coefficient K. The effective V_{max} is enhanced by K (more enzyme), and the effective [S] is enhanced by K (more substrate):

$$v_{\text{condensate}} = \frac{K\,V_{\max}\,K\,[S]}{K_m + K\,[S]}$$

At low substrate concentrations ($K\,[S] \ll K_\mathrm{m}$), the rate scales as K^2 — the full bimolecular enhancement. At saturation ($K\,[S] \gg K_\mathrm{m}$), the rate scales as K (only the enzyme enhancement matters).

5.2 Replication vs degradation

RNA is chemically unstable. In the prebiotic ocean, hydrolysis degrades RNA on timescales of hours to days. The **origin-of-life window** exists only when the replication rate exceeds the degradation rate:

$$v_{\text{rep}} > v_{\text{deg}} \iff K > K^*$$

where K^* is the critical partition coefficient. Below K*, degradation wins and no self-replicating system can persist.

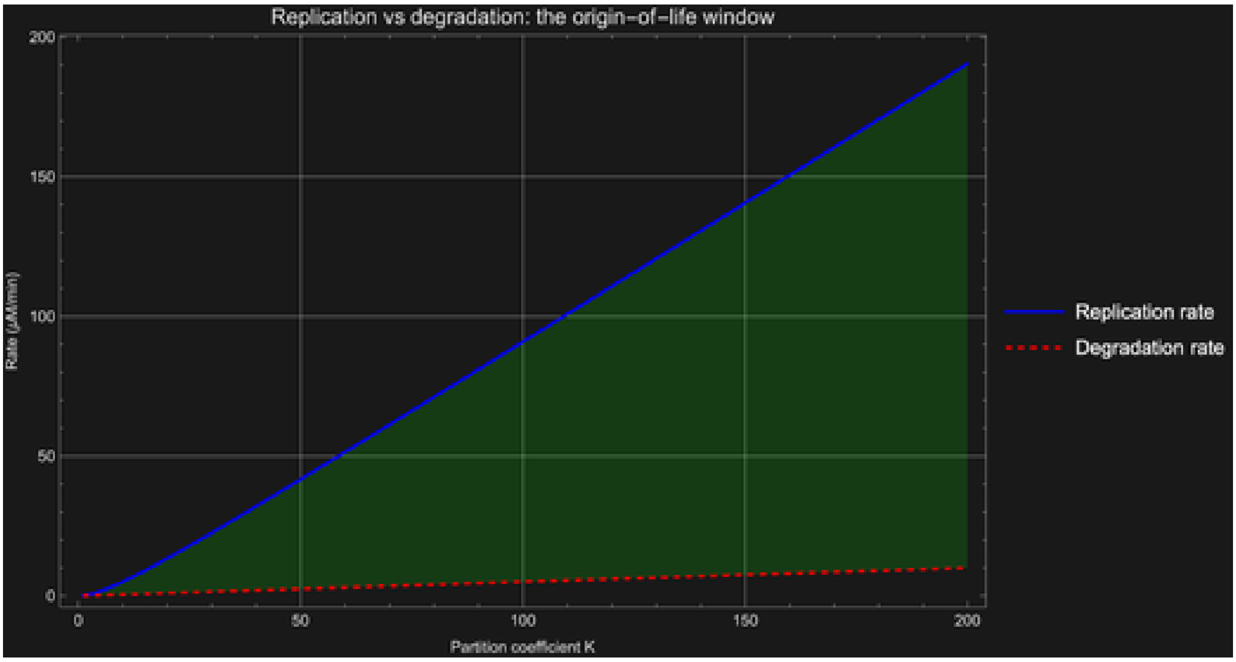


Figure 5.1 Left: Ribozyme replication rate as a function of bulk substrate concentration at different partition coefficients K. Right: Replication (blue) vs degradation (red) rate as a function of K at fixed bulk substrate [S] = 1 μM. The green-shaded region is the origin-of-life window where replication outpaces degradation.

5.3 The origin-of-life window

The critical partition coefficient K* depends on the ribozyme efficiency (V_{max}/K_m) and the degradation rate constant. For plausible prebiotic parameters ($V_{max} \approx 1\ \mu$M/min, $K_m \approx 10\ \mu$M,

k^d_{eo} ≈ 0.05/min), we find K* ≈ 20–50. Modern coacervate droplets routinely achieve K = 50–500, well above this threshold. The concentration-enhancement mechanism is not marginal — it is comfortably within the window that sustains replication.

This is the quantitative version of Oparin's coacervate hypothesis: phase separation provides the thermodynamic free lunch that bootstraps biochemistry from a dilute ocean.

6. Protocell dynamics

6.1 Growth and division

A condensate droplet that sustains net internal replication grows. Material produced inside the droplet increases its volume. When the droplet exceeds a critical radius (set by the balance between surface tension and internal compositional stress), it becomes mechanically unstable and **divides** into two daughter droplets. This is not cell division in the modern sense — there is no DNA, no cytoskeleton, no division machinery. It is purely a consequence of soft-matter physics: a liquid droplet under internal pressure that exceeds the Laplace pressure will spontaneously fission.

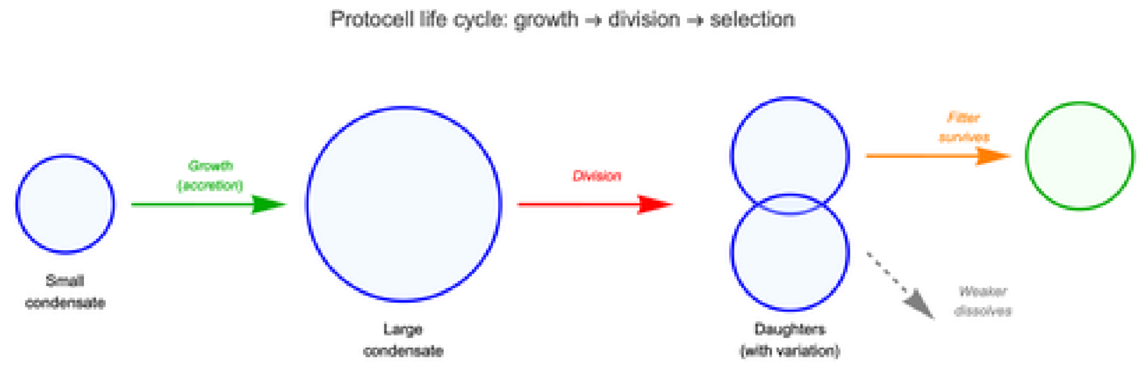


Figure 6.1 Schematic of the protocell life cycle. Small condensates grow by accretion of material from the bulk and by internal replication. When they exceed a critical size, mechanical instability drives division into two daughters. Daughters inherit (with variation) the parent's composition, including its partition coefficient K and replication efficiency. Weaker daughters dissolve; fitter ones survive and grow.

6.2 Proto-Darwinian selection

If daughter droplets inherit their parent's chemical composition (approximately), and if that composition affects their fitness (replication rate, stability), then we have the three ingredients of natural selection: **variation** (daughters differ slightly from each other), **inheritance** (daughters resemble parents more than random droplets), and **differential fitness** (some compositions grow faster than others). This is **proto-Darwinian selection** — natural selection operating on condensate populations before the invention of genetic material.

We simulate this by initialising a population of 50 protocells with random partition coefficients K and replication rates, then letting them grow, divide, compete for resources, and either survive or dissolve:

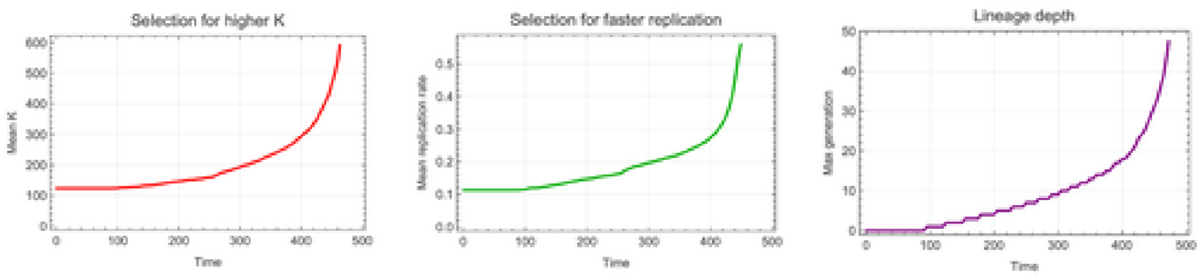


Figure 6.2 Proto-Darwinian selection in a simulated protocell population. Left: mean partition coefficient K increases over time as protocells with higher K out-compete those with lower K. Centre: mean replication rate increases. Right: lineage depth (max generation number) grows as successful lineages proliferate. These trends are the signature of adaptive evolution operating on membrane-free liquid droplets.

7. Condensates and disease

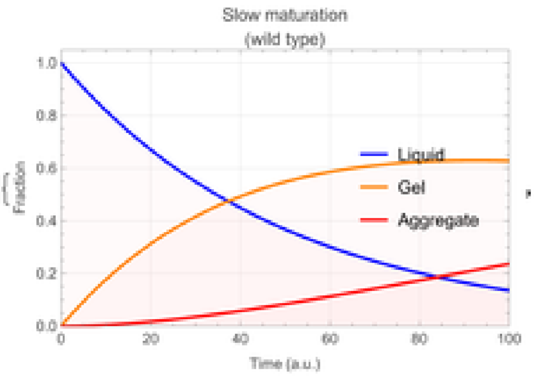
7.1 Liquid to gel to aggregate

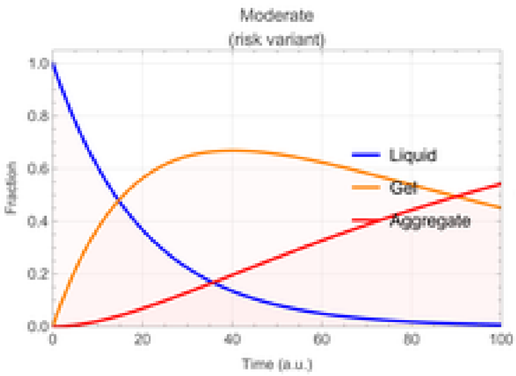
Functional condensates are **liquid**: they flow, they exchange material rapidly with the surrounding cytoplasm, and they dissolve when the cell no longer needs them. But this liquidity is not guaranteed. Over time, or in the presence of disease-associated mutations, condensates can undergo a **maturation** transition: liquid → gel → solid aggregate. The gel state has reduced dynamics; the solid aggregate is essentially irreversible.

This maturation is the molecular basis of several **neurodegenerative diseases**. The proteins FUS, TDP-43, and tau all form functional liquid condensates in healthy neurons. Disease mutations (in ALS, FTLD, Alzheimer's) shift the phase boundaries, accelerating the liquid-to-gel-to-aggregate transition. The resulting solid aggregates are the amyloid plaques and inclusion bodies seen in patient brains.

We model this as a three-state kinetic system:

$$[\text{liquid}] \xrightarrow{k_{\text{gel}}} [\text{gel}] \xrightarrow{k_{\text{agg}}} [\text{aggregate}]$$

Show[GraphicsGrid[{{ 

, 

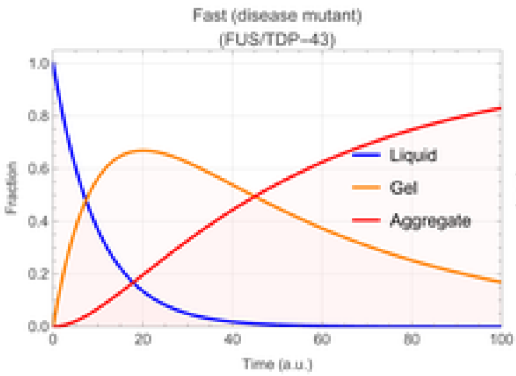
, 

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Figure 7.1 Three kinetic scenarios for condensate aging. Left: wild-type protein with slow maturation (healthy neuron). Centre: risk variant with moderate rates. Right: disease mutant (FUS/TDP-43 in ALS) with fast maturation — the liquid phase is rapidly depleted, and toxic aggregates accumulate.

7.2 Phase diagram of material states

The condensate material state depends on protein concentration and temperature (or interaction strength). We construct a schematic phase diagram showing the four regimes: mixed (one phase), liquid condensate, gel, and solid aggregate. Disease mutations effectively shift the boundaries, expanding the gel and aggregate regions at the expense of the functional liquid state.

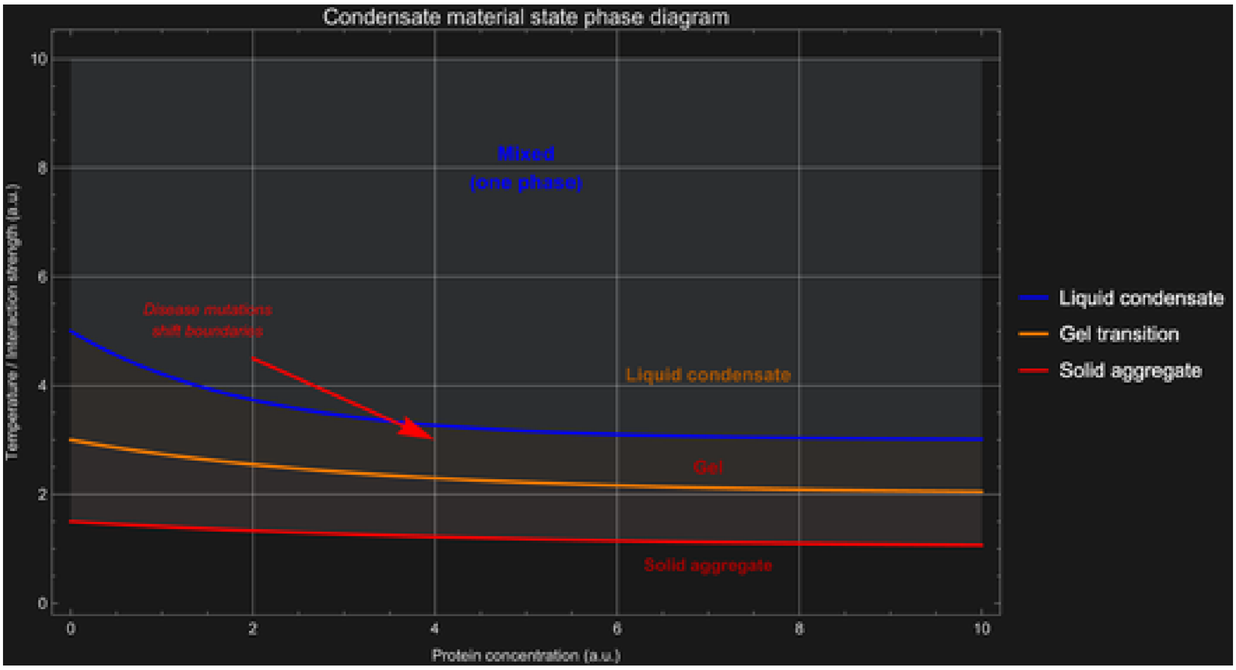

Figure 7.2 Condensate material state phase diagram. Higher concentration and lower temperature (stronger interactions) drive the progression from mixed solution to liquid condensate to gel to solid aggregate. Disease mutations shift the boundaries toward lower concentrations and higher temperatures, making pathological aggregation more likely.

8. Nonlinear dynamics: stability, bifurcation, and coarsening

The previous sections built the model; here we analyse it with the toolkit of **nonlinear dynamics** — linear stability, bifurcation theory, Lyapunov functionals, and scaling laws. Every figure in this section is generated by `wolfram/nonlinear_dynamics.wls` (with a parallel Python implementation in `src/nonlinear_analysis.py`).

8.1 Linear stability and the dispersion relation

Linearise the Cahn–Hilliard equation about a uniform state $\phi = \phi_0 + \delta\, exp(i\, k{\cdot}x + \omega t)$. Each Fourier mode grows or decays independently at the rate

$$\omega(k) = -M\, k^2 \left(f''(\phi_0) + \kappa\, k^2\right)$$

A uniform state is **unstable** precisely when $f''(\phi_0) < 0$ — the **spinodal** condition — in which case modes with $0 < k < k\_c = sqrt(-f''/\kappa)$ grow. The fastest-growing mode and its rate are $k^* = sqrt(-f''/2\kappa)$ and $\omega^* = M\,(f'')^2/4\kappa$, setting the initial droplet spacing $\lambda^* = 2\pi/k^*$. This single mode is the seed of all condensate morphology.

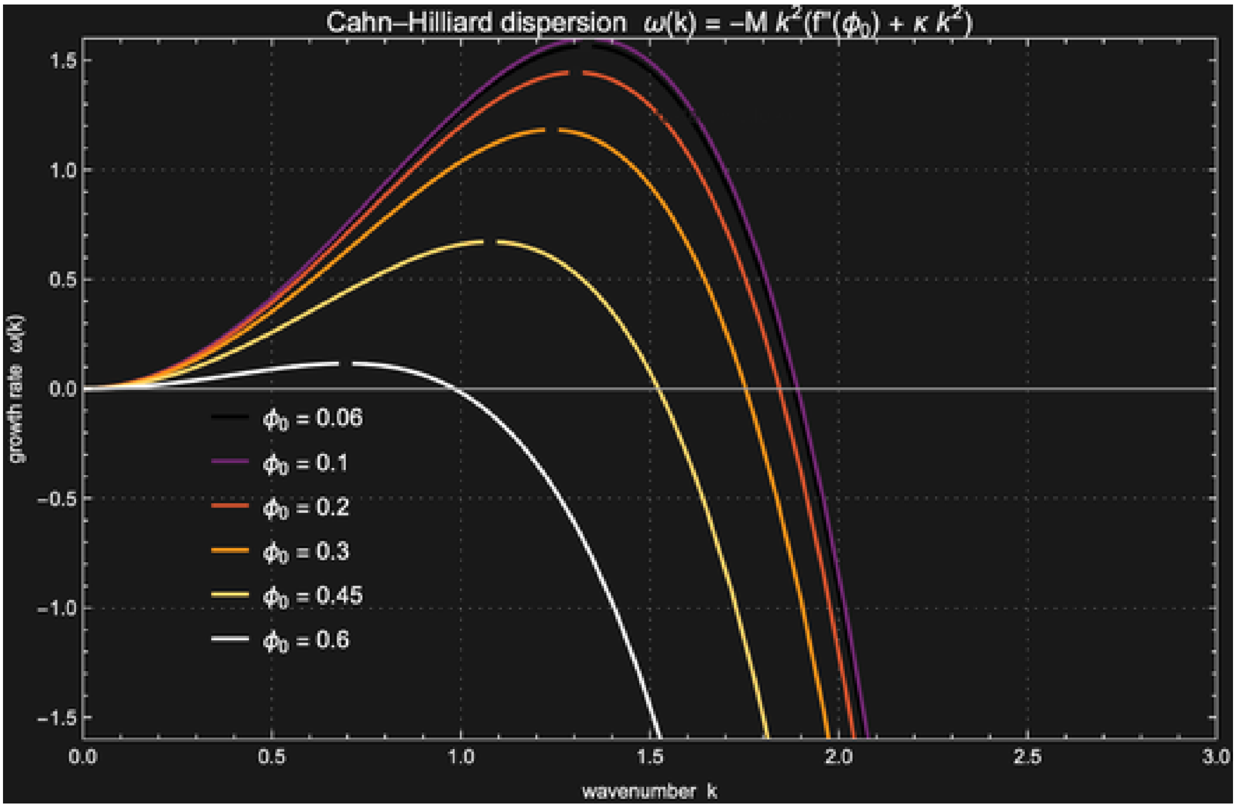

Figure 8.1 Cahn–Hilliard dispersion relation for the Flory–Huggins free energy ($\chi = 1.5$). Curves with a positive lobe are unstable; dots mark the fastest-growing mode k. As the mean composition ϕ_0 approaches the spinodal edge, the growth rate collapses to zero and the selected wavelength diverges.*

8.2 Mode selection across the parameter plane

The selected wavelength $\lambda^* = 2\pi\ sqrt(2\kappa/(-f''))$ grows with the surface-tension parameter κ and diverges at the spinodal boundary, where $f'' \to 0$. The map below shows how the length scale is selected jointly by composition and κ.

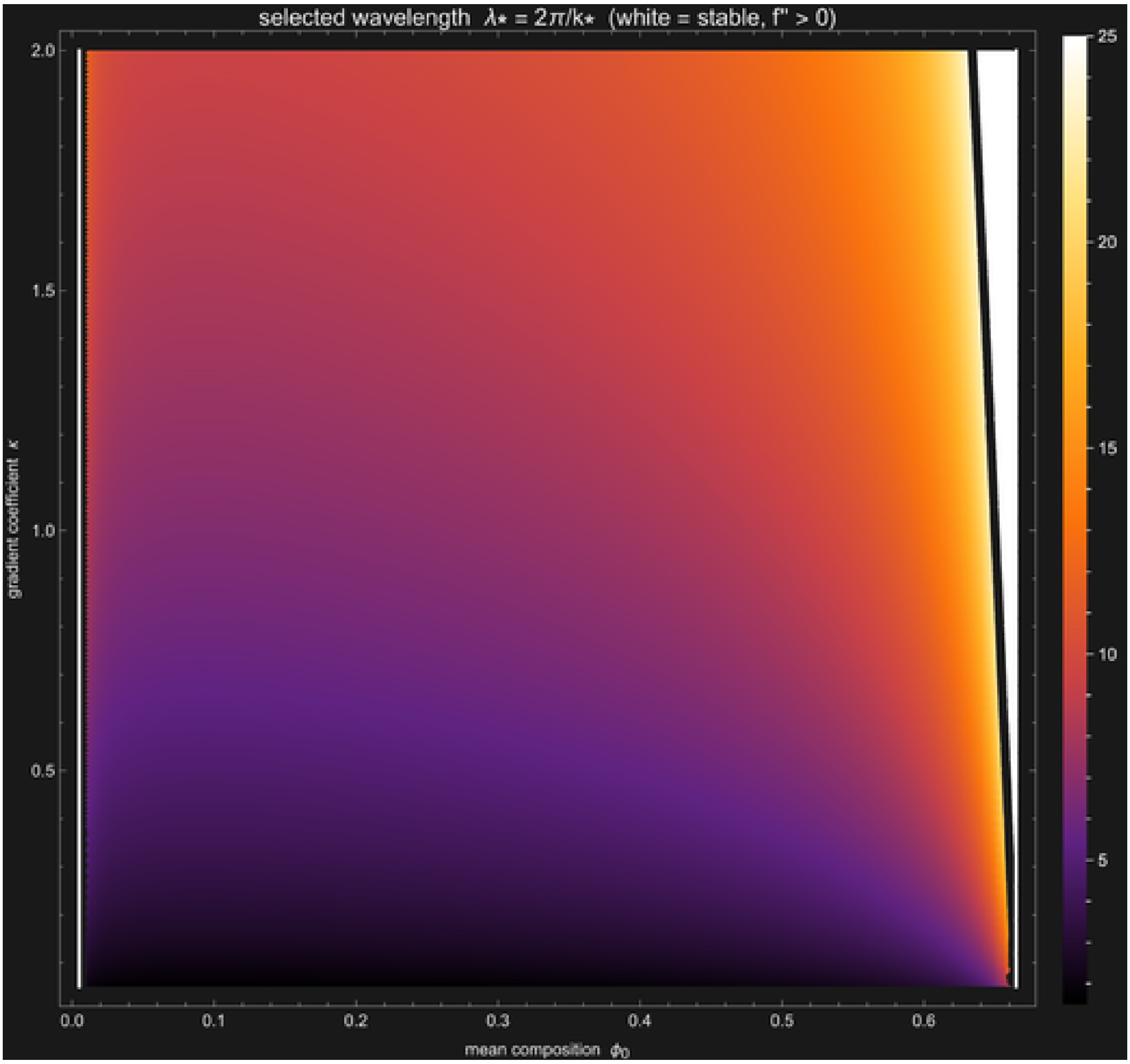

Figure 8.2 Fastest-growing wavelength λ over the (ϕ_0, κ) plane (clipped at 25 for contrast). The white curve is the spinodal $f_{\prime\prime} = 0$; outside it the homogeneous state is linearly stable and no length scale is selected.*

8.3 Bifurcation: common-tangent coexistence vs. the spinodal

Sweeping the control parameter traces out the **bifurcation diagram**. For the symmetric double well $f = -a/2\,\phi^2 + b/4\,\phi^4$ the coexistence amplitude follows a supercritical pitchfork: the binodal $\phi = \pm sqrt(a/b)$ encloses the spinodal $\phi = \pm sqrt(a/3b)$. For Flory–Huggins the coexistence curve is the **common-tangent** (Maxwell) construction. Because the osmotic pressure is linear in χ, the dense branch follows the closed form $\chi(\phi\_R) = (\phi\_R\, g(\phi\_R) - h(\phi\_R))/\phi\_R^2$, which is numerically robust where the repository's earlier two-variable solver failed.

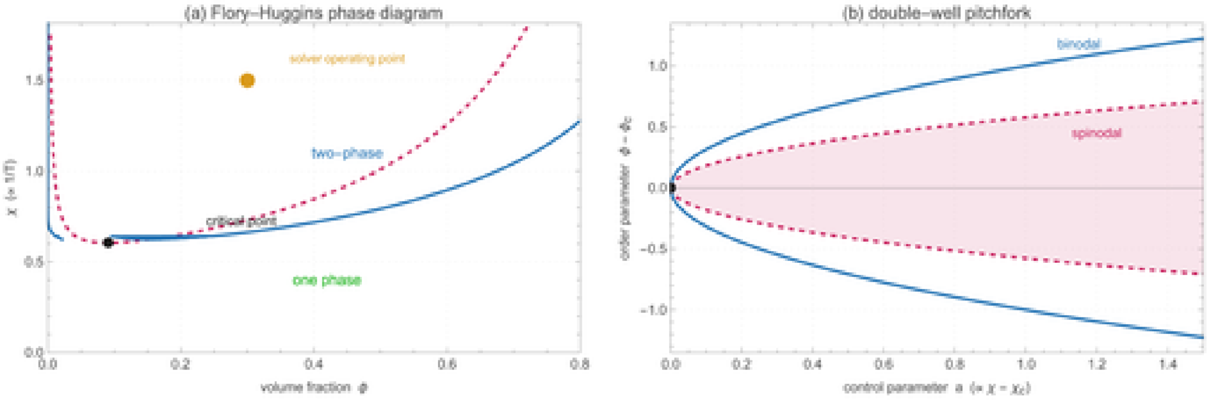

Figure 8.3 (a) Flory–Huggins phase diagram: binodal (coexistence, blue) outside the spinodal (instability, red dashed), meeting at the critical point; the orange dot marks the solver operating point ($\phi = 0.3$, $\chi = 1.5$). (b) The symmetric double-well pitchfork, with the metastable region between binodal and spinodal.

8.4 The free energy as a Lyapunov functional

Cahn–Hilliard dynamics are a **gradient flow** of the free-energy functional $F[\phi] = \int [\, f(\phi) + (\kappa/2)\, |\nabla\phi|^2 \,]\, dx$. Conservation of ϕ and the structure of the equation guarantee $dF/dt \leq 0$: F is a **Lyapunov functional** and the phase-separated state is its minimiser. Each coarsening event — two domains merging — appears as a step down in F.

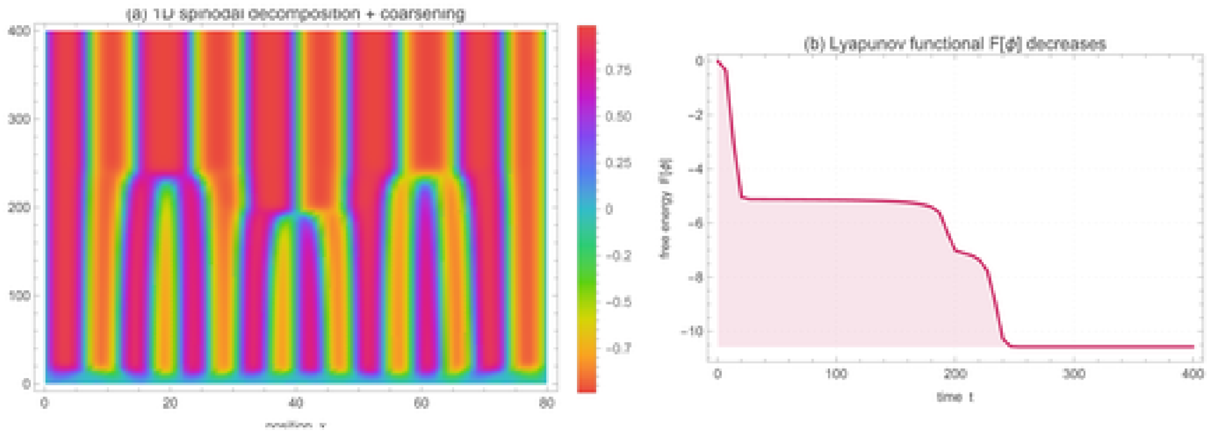

Figure 8.4 (a) Space–time evolution of the 1D double-well Cahn–Hilliard equation (NDSolve): domains form, then coarsen by merging. (b) The free energy F[ϕ] decreases monotonically, falling in discrete steps as domains coalesce — the signature of a Lyapunov functional. (Note: the strongly stabilised Flory–Huggins solver instead freezes; the canonical t^(1/3) coarsening, with $L^3 = L_0^3 + K t$ linear in time, is reproduced by the double-well Model B in the Python implementation.)

8.5 Mass-conserving reaction enhancement

Section 4 estimated the bimolecular rate boost inside a condensate as $K_A K_B$. That is the *open-system* limit (buffered reservoir). Imposing **mass conservation** — the condensate concentrates reactants but *depletes* the surrounding dilute phase — gives a very different answer. With dense area fraction p and rapid partitioning $c_{in} = K c_{out}$ in each phase, the volume-averaged rate relative to a well-mixed system is

$$\frac{k_{\text{eff}}}{k_{\text{out}}} = \frac{p\, K_A\, K_B + (1 - p)}{(1 + p\,(K_A - 1))\,(1 + p\,(K_B - 1))}$$

The enhancement vanishes at both $p \to 0$ (no condensate) and $p \to 1$ (everything uniform); it is maximal at an **intermediate dense fraction** $p^* \approx 1/K$, where the gain is only $\approx K/4$ — not the naive K^2. Bulk depletion is the hard ceiling on condensate catalysis, and it predicts an optimal droplet volume fraction — a falsifiable, quantitative claim.

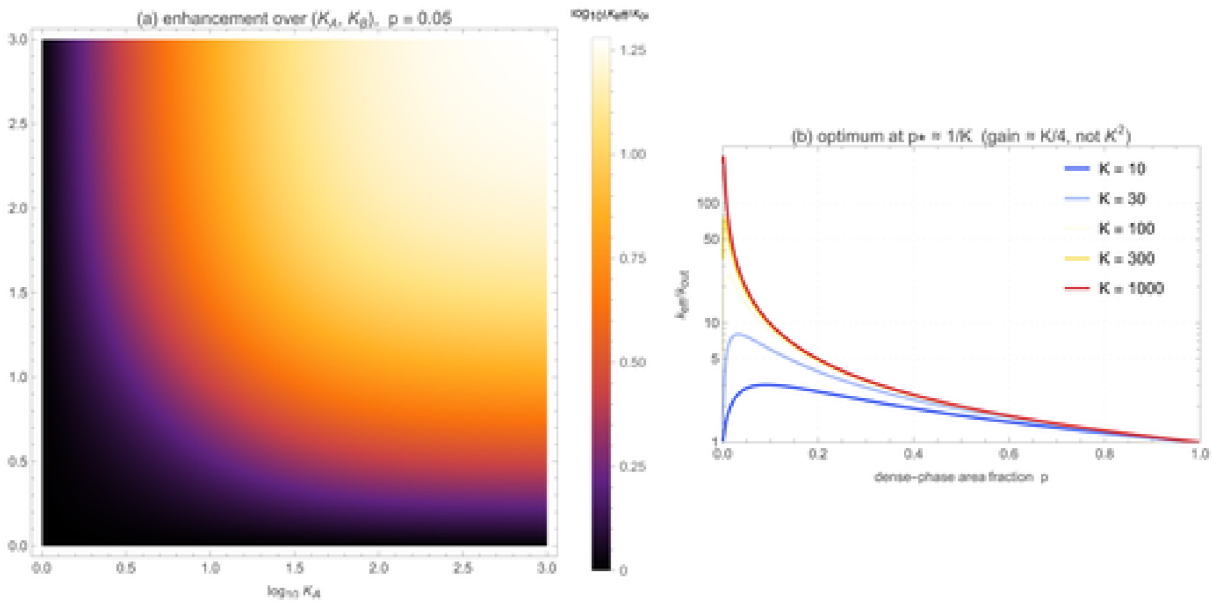

Figure 8.5 Mass-conserving rate enhancement k_eff/k_out. (a) Dependence on the partition coefficients at fixed dense fraction p = 0.05. (b) Dependence on dense fraction for several K: each curve peaks at p ≈ 1/K with a maximum gain of order K/4, far below the open-system K^2.*

9. Conclusions

What we showed. (1) The Flory–Huggins lattice theory, implemented in a few lines of Wolfram Language, correctly predicts the qualitative phase behaviour of biomolecular condensates: a UCST phase diagram with the critical concentration shifting to lower values for longer/more multivalent molecules. (2) The Cahn–Hilliard equation, solved natively by `NDSolve`, captures the dynamics of spinodal decomposition and the subsequent coarsening (Lifshitz–Slyozov t^(1/3) scaling). (3) The concentration enhancement argument (K^2 for bimolecular reactions) provides a quantitative resolution to the dilute-ocean problem in prebiotic chemistry. (4) Protocell simulations show that condensate populations can undergo proto-Darwinian selection without genetic material. (5) The liquid–to–gel–to–aggregate transition connects condensate physics to neurodegenerative disease.

The bigger picture. Biomolecular condensates sit at the intersection of soft-matter physics, molecular biology, the origin of life, and neurodegeneration. The mathematical framework — Flory–Huggins thermodynamics, Cahn–Hilliard dynamics, Michaelis–Menten kinetics, Lifshitz–Slyozov coarsening — is classical and well-understood. What is new is the recognition that this physics operates inside living cells, that it provides the thermodynamic foundation for protocellular compartments, and that its failure modes are disease. The condensate revolution is not a discovery of new physics; it is the discovery that old physics was hiding in plain sight inside biology.

10. References

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11. Reproducibility

All source code lives in the repository **github.com/mthiel74/biomolecular-condensates** (`https://github.com/mthiel74/biomolecular-condensates`). The pipeline has two parallel tracks:

Wolfram Language (analysis scripts in `wolfram/` and this notebook builder in `community/`) and Python (`src/`, including a full 2D Cahn–Hilliard spectral solver). To regenerate this notebook from scratch:

```
# 1. Run all Wolfram analysis scripts (generates docs/images/*.png)
wolframscript -file wolfram/run_all.wls

# 2. Rebuild this notebook
wolframscript -file community/build_notebook.wls

# Or, for the Python track:
pip install numpy scipy matplotlib
python3 src/flory_huggins.py
python3 src/cahn_hilliard.py
python3 src/condensate_kinetics.py
python3 src/protocell.py
python3 src/nonlinear_analysis.py   # nonlinear-dynamics analysis (§8)
```

Generated on June 20, 2026 with Wolfram Language 15..